

GENERALIZED COSET CONSTRUCTION AND CONSTRAINED FERMIONIC MODELS

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We present a path-integral formulation of two-dimensional constrained fermionic models which correspond to a generalized coset construction $G = H_n/H_{n-1}/\dots/H_1$ for a descending chain of groups $H_n \supset H_{n-1} \supset \dots \supset H_1$.

Examples of the coset construction already appeared in the early period of string theory.^{1,2} More recently, Goddard, Kent and Olive (GKO)³ presented an algebraic formulation of the coset construction which provided a systematic approach to rational conformal field theories. This construction led to new insights on the properties of models in the conformal and superconformal discrete series.⁴

Already in Ref. 3 GKO suggested that their algebraic construction should be realized by some fermionic model with symmetry group G from which the degrees of freedom of a subgroup $H \subset G$ are removed. This process is in fact reminiscent of the definition of a coset space G/H . It is important to note that the energy-momentum tensor of the resulting G/H model obeys a Virasoro algebra with a conformal charge

$$c_{G/H} = c_G - c_H. \quad (1)$$

Field theoretical realizations of this construction have been presented using gauged bosonic^{5,6} and fermionic^{7,8} models. In this way, the principal series, superconformal series, etc. were obtained in a path-integral Lagrangian approach. Much can be learned about coset models by realizing them as gauged Wess-Zumino-Witten or gauged fermionic models (see for example Ref. 9 and references therein).

Recently, Witten¹⁰ has observed that coset models have a generalization to an arbitrary descending chain of compact groups $H_n \supset H_{n-1} \supset \dots \supset H_1$ (the original coset construction referred to above corresponds to the $n = 2$ case). Since there is no natural notion of such an n -step coset space $G = H_n/H_{n-1}/\dots/H_1$ from the

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mathematical point of view, it is interesting to analyze, using field theoretical realizations, the resulting conformal models (Witten construction of these Generalized Coset (GC) models is purely algebraic).

In the present work we present an explicit field theoretic realization of GC models in terms of two-dimensional constrained fermionic models. Apart from providing an intuitive understanding of the GC models, our construction should allow, in a Statistical Mechanics context to a thorough analysis on how the inclusion of relevant operators perturbs a system away from the critical point.

We mainly follow the approach presented in Ref. 8. Basically we add gauge fields to an otherwise free fermionic Lagrangian. These gauge fields act as Lagrange multipliers which realize the desired constraints. We find that the resulting GC models have a central charge given by a formula which generalizes (1) in a natural way. Concerning the Kac-Moody (KM) algebra associated to each subgroup, its central charge can also be easily computed and depends on the different subgroups of the chain. For example, if one starts from a Lagrangian for free left-handed fermions in the fundamental representation of $G = U(n)$ ($n = N \times k$) and then one takes a subgroup $H = SU(k)$ and some subgroup $L (L \subset H)$, the resulting $G/H/L$ model has a central charge $c_{G/H/L}$ given by

$$c_{G/H/L} = c_{G,1} - c_{H,k} + c_{L,k+2(C[H]-C[L])}. \quad (2)$$

On the right-hand side we indicate with $c_{A,p}$ the Virasoro central charge associated with a group A and a KM level p . We denote by $C[A]$ the Casimir operator for A in the adjoint representation.

We end this work with a discussion of more general cases and their connection with discrete series.

Let us consider n free left-handed fermions in the fundamental representation of $G = U(n)$, with Lagrangian (in two-dimensional Euclidean space):

$$\mathcal{L}_0 = \psi_L^\dagger i \partial_- \psi_L. \quad (3)$$

We take $n = N \times k$ thus distinguishing a “color” index running from 1 to N from a “flavor” index running from 1 to k .

Were we interested in the usual G/H coset construction, we should introduce at this stage^{7,8} gauge fields in the Lie algebra of some subgroup H ($H \subset G$, with generators T^a , $a = 1, 2, \dots, \dim H$). These gauge fields should act as Lagrange multipliers enforcing the constraints:

$$J^a |\text{phys}\rangle = 0 \quad (4)$$

so that physical states $|\text{phys}\rangle$ are singlet under the “flavor” currents J^a ,

$$J^a = \psi_L^\dagger T^a \psi_L. \quad (5)$$

In this way, one arrives⁸ to an effective action which corresponds to a $U(N \times k)/H$ coset model with Virasoro charge given by (see (1)):

$$c_{U(n)/H} = c_{U(n)} - c_H. \quad (6)$$

This result can be easily generalized to an arbitrary group G with a level k KM algebra.

Now, suppose we do not stop at this stage but continue one more step in constraining the model. That is, instead of considering the whole subgroup H , we constraint currents belonging to the coset H/L (with generators T^i , $i = 1, 2, \dots$, $\dim H - \dim L$), where L is a subgroup of H . The generating functional associated with the resulting $G/H/L$ GC model reads:

$$Z_{G/H/L} = \int D\psi_L^+ D\psi_L DA_-^a \exp(-\mathcal{L}) \prod_{i=1}^{\dim L} \delta[A_-^i]. \quad (7)$$

In Eq. (7) the gauged fermionic Lagrangian $\mathcal{L}$,

$$\mathcal{L} = \psi_L^+ (i\partial_- + A_-) \psi_L \quad (8)$$

has been constructed from $\mathcal{L}_0$ after the use of the identity

$$\prod_{i=1}^{\dim H} \delta[J_+^a] = \int DA_-^a \exp\left(-\int d^2x J_+^a A_-^a\right), \quad (9)$$

which would correspond to constraining all currents in the subgroup H . Precisely δ -functionals on the right-hand side of (7) have been included so as to ensure that only currents on H/L are effectively constrained. We proceeded in this way because the fermionic path-integral in (7) can be very easily evaluated. Indeed, it reduces to a (chiral) two-dimensional determinant which can be related to the ordinary (Dirac) determinant in the $A_+ = 0$ gauge.¹¹ That is, being chiral determinants ill-defined quantities, we add free fermions of the opposite chirality which do not couple to physical states. We shall then adopt the following definition:

$$\begin{aligned} & \int D\psi_L^+ D\psi_L \exp(\psi_L^+ (i\partial_- + A_-) \psi_L) \\ & \equiv \frac{1}{\det i\partial_+} \int DA_+ \det(i\partial + \not{A}) \delta[A_+] \Delta_{\text{FP}}. \end{aligned} \quad (10)$$

On the right-hand side of (10) we are working with (two-component) Dirac fermions in the axial ($A_+ = 0$) gauge. In this gauge, only left-handed components are coupled. The Faddeev-Popov determinant Δ_{FP} ensures the correct normalization for the identity and the (free-fermion) determinant in the denominator cancels the effect of spurious right-handed degrees of freedom.

Now, the Dirac determinant in (10) can be written in terms of the Wess-Zumino-Witten (WZW) action for a H -valued field¹¹ h

$$A_- = \frac{1}{i} h^{-1} \partial_- h. \quad (11)$$

We then have

$$\int D\psi_L^\dagger D\psi_L \exp(\psi_L^\dagger (i\partial_- + A_-) \psi_L) = \det i\partial_- \exp(-kW[h]) \Delta_{\text{FP}} \quad (12)$$

with

$$W[h] = \frac{1}{8\pi} \text{tr} \int d^2x \partial_\mu h^{-1} \partial_\mu h + \text{tr} \frac{1}{24\pi} \int d^2x \varepsilon_{ijk} h^{-1} \partial_i h h^{-1} \partial_j h h^{-1} \partial_k h \quad (13)$$

and

$$\Delta_{\text{FP}} = \int D\eta^{a-} D\varepsilon^a \exp \int d^2x \eta^{a-} \partial_+ \varepsilon^a \quad (14)$$

with ghosts η^{a-} , ε^a in the adjoint representation of H . We also use¹¹

$$DA_- = \exp(-2C[H]W[h]) Dh \quad (15)$$

with $C[H]$ the Casimir operator in the adjoint representation of H .

We finally have

$$Z_{G/H/L} = \det i\partial_- \Delta_{\text{FP}} \int Dh \exp(-(k + 2C[H])W[h]) \times \prod_{i=1}^{\dim L} \delta[h^{-1} \partial_- h|_i]. \quad (16)$$

With $h^{-1} \partial_- h|_i$ we indicate the projection of the H -valued $h^{-1} \partial_- h$ element onto the L subalgebra. We are then faced in Eq. (16) to constraining a WZW model. To this end, we introduce new Lagrange multipliers B_+^i in order to represent the δ -functionals:

$$\begin{aligned} Z_{G/H/L} &= \det i\partial_- \Delta_{\text{FP}} \int Dh DB_+^i \\ &\times \exp(-(k + 2C[H])W[h] + \text{tr} \frac{1}{4\pi} \int d^2x B_+ h^{-1} \partial_- h) \end{aligned} \quad (17)$$

with $B_+ = B_+^i t^i$ ($i = 1, 2, \dots, \dim L$).

If we now write

$$B_+ = \frac{1}{i} b^{-1} \partial_+ b \quad (18)$$

and use the Polyakov-Wiegmann identity¹¹

$$W[hb^{-1}] = W[h] + W[b^{-1}] + \frac{1}{4\pi} \text{tr} \int d^2x b^{-1} \partial_+ b h^{-1} \partial_- h, \quad (19)$$

we have

$$Z_{G/H/L} = \det i\partial_- \Delta_{\text{FP}} \int Dh \exp(-(k + 2C[H])W[h]) \\ \times \int Db \exp((k + 2(C[H] - C[L])W[b]) . \quad (20)$$

In order to compute the Virasoro central charge associated to the model described by (20) we observe that:

(i) free left-handed fermions contribute with

$$c_{\text{free fermions}} = k \times N . \quad (21)$$

(ii) There is a (-2) contribution to the central charge for each one of the $\dim H$ ghosts introduced to represent the Faddeev-Popov determinant:

$$c_{\text{ghosts}} = -2 \dim H . \quad (22)$$

(iii) Fields h and b with WZW actions $W[h]$ and $W[b]$ contribute to the central charge as follows¹²

$$c_{H,(k+2C[H])} = -\frac{(k + 2C[H]) \dim H}{-(k + 2C[H]) + C[H]} \quad (23)$$

$$c_{L,(k+2(C[H]-C[L]))} = \frac{(k + 2(C[H] - C[L])) \dim L}{k + 2(C[H] - C[L]) + C[L]} . \quad (24)$$

Putting all this together, we have

$$c_{G/H/L} = k \times N - \frac{k \dim H}{k + C[H]} + \frac{(k + 2(C[H] - C[L])) \dim L}{k + 2(C[H] - C[L]) + C[L]} , \quad (25)$$

an expression which can be written in the announced form (2)

$$c_{G/H/L} = c_{G,1} - c_{H,k} + c_{L,k+2(C[H]-C[L])} . \quad (26)$$

We then see that the central charge of a GC model can be expressed in terms of sums and differences of central charges associated to each subgroup, a result which could be guessed without having an explicit field realization as the one we have given. Less obvious is the resulting level for the KM algebra associated with each Virasoro central charge. Equation (26) can be easily extended to a general chain $G/H_{n-1}/H_{n-2}/\dots/H_1$ with a value of c :

$$c_{G/H_{n-1}/H_{n-2}/\dots/H_1} = c_{G,1} - c_{H_{n-1},k_{n-1}} + \dots \pm c_{H_1,k_1} . \quad (27)$$

The resulting KM levels k_i ($i = 1, 2, \dots, n-1$) can be easily calculated and depend on the Casimir of different subgroups.

It is easy to check that when H is a direct product of two groups, $H = H_1 \otimes H_2$ and L coincides with one of the factors (say H_2), the central charge of the resulting GC model is:

$$c_{G/H_1 \otimes H_2/H_2} = c_{G,k} - c_{H_1,k} \quad (28)$$

a result that could be advanced from the GKO construction.³

An interesting application of our GC construction corresponds to the case $G = \text{SU}(2)_k \otimes \text{SU}(2)_1$, $H = \text{SU}(2)_{k+1}$ and $L = \text{U}(1)$ (with subindices we indicate KM levels of different algebras). Note that for this choice of G and H , GKO³ have shown that $c_{G/H}$ corresponds to the FQS series. In order to obtain the GC model we proceed as before. That is, we start from a fermionic G/H model with additional constraints

$$\begin{aligned} Z_{G/H/L} = & \int D\psi_L^\dagger D\psi_L D\chi_L^\dagger D\chi_L DB_+ DA_+ Db_+ Da_+ \delta[J^3] \\ & \times \exp \left(- \int d^2x [\psi_L^\dagger (i\partial_+ + A_+ + B_+ + b_+) \psi_L \right. \\ & \left. + \chi_L^\dagger (i\partial_+ + A_+ + a_+) \chi_L] \right), \end{aligned} \quad (29)$$

where $B_+ = B_+^a T^a \times I$ and $A_+ = A_+^i I \times t^i$, (T^a , t^i are the generators in the fundamental representation of $\text{SU}(k)$ and $\text{SU}(2)$ respectively) and a_+ , b_+ are $\text{U}(1)$ gauge fields. The δ -functional on the right-hand side of (29) enforces the $L = \text{U}(1)$ constraint.

The Virasoro central charge for the resulting model can be easily calculated taking into account, as before, free fermions, ghosts and WZW contributions. The answer is

$$c_{G/H/L} = 2 - \frac{6}{(k+2)(k+3)}. \quad (30)$$

Again we have

$$c_{G/H/L} = c_G - c_H + c_L. \quad (31)$$

The models with values of c in this new series (which does not coincide with any of the previously known) are unitary but non-minimal. Analogous constructions can be done in connection with superconformal series, parafermionic series, etc.

In summary, we have been able to construct constrained fermionic models which provide a field-theoretic realization of the construction proposed in Ref. 10 as a generalization of coset models. Many aspects of the models we have presented remain to be investigated: in particular, how the discrete series arising from the G/H coset construction could be modified increasing the chain of constraining subgroups. Also, the analysis of GC models in the canonical approach, taking into account the second class constraints that may arise for certain chains of H_i 's subgroups should

complement our path-integral formulation. Work on these aspects will be presented elsewhere.

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